Oxytocin

Mechanism of action

- Oxytocin is a synthetic version of the naturally occurring hormone oxytocin which is normally released from the pituitary gland.
- Oxytocin stimulates oxytocin receptors on the uterus, causing increased uterine contraction and reducing blood loss from the uterus.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Following normal birth.
- ✓ Postpartum haemorrhage.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- For following normal birth: 10 units IM.
- For postpartum haemorrhage:
  - 10 units IV (preferred), or
  - 10 units IM.

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- If oxytocin IM has already been administered as part of routine treatment following normal birth, an additional dose of oxytocin should be administered (in the other thigh if administering IM) if postpartum haemorrhage develops. This may require meeting another vehicle.

Administration

- IV administration:
  - Place 10 units into a 1 litre bag of 0.9% sodium chloride. This solution contains 0.01 units/ml.
  - Shake well and label.
  - Administer IV over approximately 60 minutes (approx. 4 drops per second).
  - The rate of administration may be increased if shock is present.
- IM administration:
  - Administer IM undiluted.
  - The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.
  - If multiple babies are present, administration must occur after delivery of the last baby.

Common adverse effects

- Abdominal cramping.
- Tachycardia.
- Flushing.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 30-60 minutes.

Usual preparation

- Ampoule containing 10 units in 1 ml.

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Pharmacokinetics

- Oxytocin is metabolised in the liver and kidneys.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

- None.

Additional information

- Oxytocin that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is eight weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Routine administration of oxytocin following normal birth is controversial, but appears to reduce the incidence of postpartum haemorrhage.
- Oxytocin has been reported to cause prolongation of the QT interval if the patient is taking other medicines that also prolong the QT interval. However, this is usually associated with prolonged IV infusions and is not a clinically significant consideration when administering one or two IM doses.

More Information

Version history

9 June 2025 1.0.5.3

Update: Dosage and administration updated.